Development of an ANN-Based Diagnostic Model for Yellow-Flesh Watermelon Ripeness Classification Using VIS and NIR Spectral Properties with Various Training Functions

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Abstract — **Traditional methods for determining watermelon ripeness through destructive or manual tapping are subjective and not suitable for large-scale grading. This study developed an intelligent diagnostic model using ANN to categorize the grades of the yellow-fleshed watermelons ripeness according to the optical reflectance properties of the Visible Light Spectrum, VIS at 751nm and Near-Infrared, NIR at 1058nm. The data were organized into four parameters to represent the two ripeness grades which are grades A and B, while the data was analyzed initialized using basic descriptive statistical analysis using mean, minimum, maximum and standard deviation. Three training functions based on mean reduction in MATLAB (traingd, traingdm and traingda) were evaluated across 14 hidden neuron configurations. The outcome shows that the traingda model with 21 hidden neurons performed best with 86.7% sensitivity, 80.0% specificity and 83.3% accuracy. The results demonstrated that the ANN with adaptive learning rate algorithm offered a betternon-destructive method for determining the quality of watermelons with yellow flesh.**

Keywords — Artificial Neural Networks; watermelon ripeness; VIS/NIR spectroscopy; non-destructive classification; training function; MATLAB

I. INTRODUCTION

One of the most popular fruits in tropical countries like Malaysia is watermelon, also known as Citrullus lanatus, which is valued for its sweetness, freshness of the fruit and high-water content in the fruit [4], [5]. Yellow-fleshed varieties have gained high commercial demand due to their distinctive color and taste. For consumers and producers, the internal ripeness of the fruit is a key variable in determining the quality of watermelon consumed by humans, which is closely related to the soluble solids content and flesh development [9], [13]. Therefore, the ripeness grade directly affects the market value of the fruit, and reliable grading is essential for quality control and consumer satisfaction [1]. However, because watermelons have thick and opaque skins, their internal condition cannot be directly observed, and the fruit cannot be well assessed from external appearance alone [4]. Conventionally, watermelon ripeness is assessed using destructive or subjective methods [2,3]. Destructive methods require cutting the fruit to inspect the flesh or to measure the sugar content using a refractometer, which makes the sample unmarketable [9]. Subjective methods rely on tapping or banging the fruit and interpreting the resulting sound, a skill that is entirely dependent on the experience of the fruit assessor or seller and is therefore difficult to reproduce consistently [3,15]. This is even more problematic when grading must be carried out on a large scale, where speed, consistency and absence of damage are essential [16]. With a more precise mechanism for classifying the ripeness grade of watermelons, it can be repeated and does not damage the required fruit [1,5].

The visible (VIS) and near-infrared (NIR) portions of the electromagnetic spectrum, which range from approximately 380 to 2500 nm, interact with the major structural and functional groups of organic compounds to produce a reflectance spectrum that provides details about the internal quality of the fruit [1]. Since these spectra can be obtained without cutting the sample, VIS and NIR spectroscopy have become an attractive basis for non-destructive quality assessment and are widely used in multispectral and hyperspectral imaging of produce [6,21,12]. Artificial neural networks (ANNs) and fuzzy logic are two intelligent techniques that have been applied to fruit quality classification in order to analyze such spectral data. [17,10]. ANNs are well suited for this study because they can learn the nonlinear relationship between optical input features and ripeness classes directly from the data, without the need for an explicit physical model [1,10].

However, the number of hidden neurons in the network and the training function used to update the weights have a significant impact on their performance. Therefore, the goal of this study is to create an ANN-based diagnostic model that uses the VIS and NIR reflectance characteristics of yellow-fleshed watermelons to classify their maturity grades., comparing three gradient-descent-based training functions, which are *traingd* namely gradient descent, *traingdm* namely gradient descent

with momentum and *traingda* namely gradient descent with adaptive learning rate, by evaluating their percentage of sensitivity, specificity and accuracy respectively [1]. Finally comparing the three gradient-descent-based training functions with models used in previous research.

II. METHODOLOGY

For Figure 1 shows a flowchart that summarizes the complete project process. The study began with data collecting, followed by data acquisition and pre-processing, basic descriptive statistics of optical qualities for two grades of yellow-fleshed watermelon (grades A and B), and lastly the development and assessment of the ANN classifier system. The performance for each training function was compared to identify the train function that yielded the highest sensitivity, specificity, and accuracy.

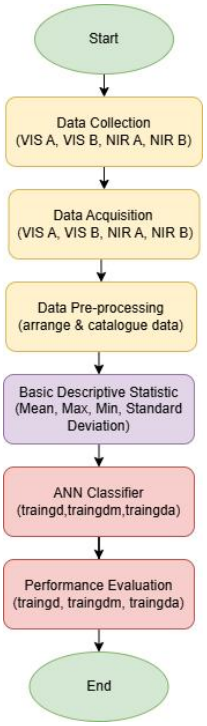


Figure 1: Methodology flowchart

A. Data collection

The optical data utilized in this study included VIS and NIR reflectance measurements obtained from uncut (non-destructive) yellow-fleshed watermelon samples [1]. The samples were divided into two maturity grades, denoted as Grade A and Grade B, distinguished by their flesh condition and weight range. The frequency of watermelon raw samples is tabulated in Table I. For each sample, reflectance was recorded in the VIS and NIR regions, and the wavelength of interest was chosen at the point of maximum reflectance for most samples [1]. At this stage, each raw sample will be divided into several partitions which accumulated 200 for each parameter. The VIS property was taken at a wavelength of 751 nm and the NIR property at 1058 nm, consistent with the wavelengths reported to give the strongest response for the optical properties of watermelon [9,13]. The average reflectance for each grade in the VIS and NIR regions is summarized in Table II.

TABLE I. FREQUENCY OF WATERMELON SAMPLES

Type of Flesh	Weight (kg)	Sample (n)	Grade
Yellow	≥ 6.1	56	A
Yellow	4.4 – 6.0	58	B

TABLE II. AVERAGE REFLECTANCE IN VIS AND VIR

Grade	VIS (%R)	NIR (%R)
A	61.49	44.40
B	65.82	36.27

B. Data acquisition and pre-processing

The reflectance data obtained from the previous research project samples were re-arranged and catalogued based on the watermelon's grade and optical characteristics [1]. This resulted in four parameters, namely VIS A and B, also NIR A and B. A total of 800 samples were from all parameters to form the data set used in subsequent analysis and training. During pre-processing, the spectral color information was examined through superimposed reflectance plots for the two grades. Figure 2 shows the multiple overlays of reflectance plots in the VIS region, while Figure 3 shows the corresponding plots in the NIR region. The selected wavelengths of 751 nm and 1058 nm are shown, as these correspond to the maximum reflectance points for most samples and provide the clearest separation between grades [1,9].

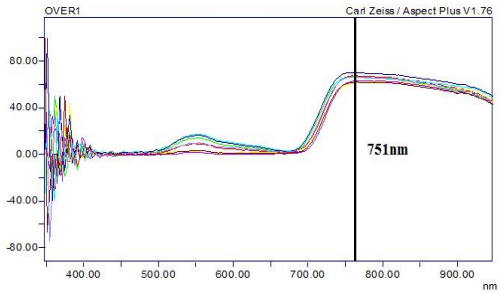


Figure 2: Overlay reflectance plot in the VIS region for yellow-flesh watermelon from previous research project.

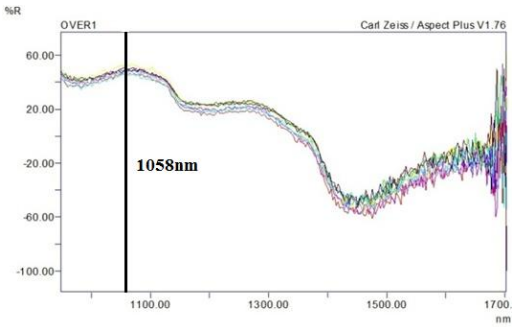


Figure 3: Overlay reflectance plot in the NIR region for yellow-flesh watermelon from previous research project.

C. Basic Descriptive Statistic

Basic descriptive statistics were used as a preliminary analysis to summarize the main characteristics of the collected yellow flesh watermelon data set [10,13]. This process involved analyzing the data distribution for physical parameters and spectral properties (VIS and NIR) to ensure that the sample covered a sufficient range of diversity for the artificial neural network (ANN) training phase [11,12].

1) Arithmetic Mean

The arithmetic mean is the average of the numerical values in the data set that serves as an indicator of central tendency for watermelon quality characteristics [11]. In this study, it was used to determine the average value of the soluble solids content (SSC) or spectral reflectance intensity to differentiate the profiles between A and B ripeness grades [10,13].

2) Minimum

The minimum value represents the lower limit or lowest data point recorded for each variable in the sample set [11]. The identification of this minimum value is important to establish the boundaries of the maturity range and ensure that the ANN model is able to detect fruit at the lowest quality level in the specified grade category [13].

3) Maximum

The maximum value refers to the highest measurement or upper limit recorded for a parameter such as the physical size or sweetness value of watermelons [11]. By knowing this maximum value, the full capacity of variation in the sample population can be determined, thus assisting in the data scaling process before input into the ANN model [13].

4) Standard Deviation (SD)

The SD is applied to determine the level of dispersion or variation of data points around the mean value [11]. A significant SD value indicates the existence of clear differences between samples of different maturity, which is necessary for assessing the constructed ANN classification model's robustness and generalizability[13].

D. Classifier system ANN

The Artificial Neural Network (ANN) classifier system process provides a summary flowchart in Figure 4 below. That classifier was developed using MATLAB using a feedforward neural network [1,10]. ANN is used to categorize watermelon ripeness based on VIS/NIR spectral data. The network is trained with the corresponding input and output data to learn patterns and make accurate predictions.

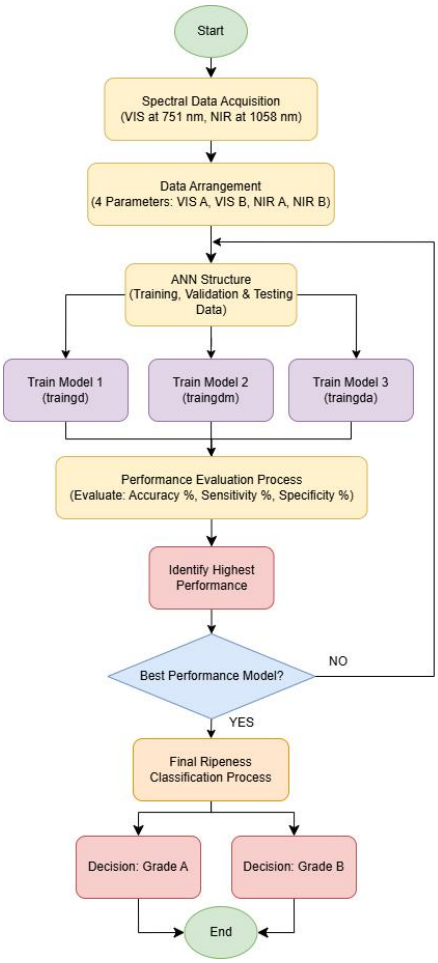


Figure 4: Flowchart of ANN Specific Classifier System

1) Data structure

A data structure that separates data into sets for training, testing, and validation. The training model's performance is accessed via the validation set. [17], the testing set is used to prevent overfitting through cross-validation, and the model parameters are trained or optimized using the training set.[1]. The development's data structure of the ANN model is presented in Table III. The complete data set is divided into training (560 samples), validation (120 samples), and testing (120 samples), corresponding to 70%, 15% and 15% respectively. The detailed sample allocation after sorting is shown in Table IV, where samples 1–280, 281–340, and 341–400 are assigned for both VIS and NIR spectral data, to train, validate, and test data sets, respectively. This data separation is necessary to train and test fairly to obtain accurate results.

The architecture network for this research uses a 2–H–1 structure with two input nodes representing each grade A and B's VIS and NIR spectra, one hidden layer containing 21 neurons (ANN Model), and one output node for watermelon ripeness classification (Yellow A or Yellow B). 14 ANN model trains with different hidden neuron configurations were evaluated to identify the optimal architecture.

TABLE III. SEPARATED DATA BEFORE SORTING

Spectrum colour code	Data Structure (IN ACTUAL NUMBERS)		
	Training	Validation	Testing
VIS Yellow A & VIS Yellow B	280	60	60
NIR Yellow A & NIR Yellow B	280	60	60
total	560	120	120

TABLE IV. SEPARATED DATA AFTER SORTING

Colour spectrum	Samples		
	1-280	281-340	341-400
VIS	A + B	A + B	A + B
NIR	A + B	A + B	A + B
MATLAB STRUCTURE	TRAINING (280)	VALIDATION (60)	TESTING (60)

2) Training algorithm

Three gradient descent-based learning algorithms were implemented and compared, which are gradient descent, gradient descent with momentum and gradient descent with adaptive learning rate [1]. The input to each network was VIS and NIR data, and the target was watermelon grade. For each training function, fourteen networks with different numbers of hidden neurons (3, 5, 7, 9, 11, 15, 17, 21, 23, 25, 27, 33, 37 and 50) were trained so that the best configuration for each algorithm could be identified.

Figure 5 shows the architecture of a feedforward ANN network developed using MATLAB, this architecture was chosen because of its ability to model the nonlinear relationship between spectral parameters and ripeness grade with a simple network structure. Figure 6 illustrates the ANN model's training method using the MATLAB Neural Network Toolbox., which performs a Gradient Descent-based training function (traingda) until the validation stopping criterion is met. This mechanism helps to avoid overfitting and improves the model's potential to produce more precise classifications on data that was never used during training.

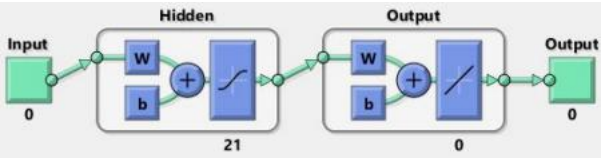


Figure 5 : Feed-forward ANN architecture (2–H–1) from MATLAB used for grade classification.

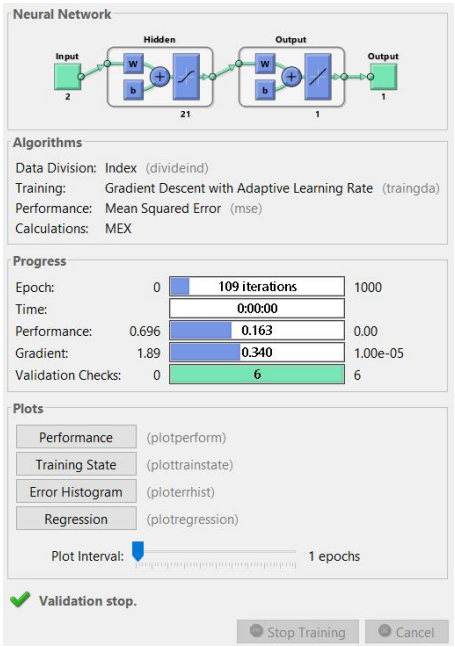


Figure 6 : Neural Netwotk Training (nntraintool) from MATLAB

3) Performance evaluation

The performance for each training network has been determined using the confusion matrix as shown Figure 7 below terms TP-true positive, TN-true negative, FP-false positive, FN-false negative and NOC-Number of Connections [1], [35]. From these terms, sensitivity, specificity and accuracy were calculated using (1), (2), (3) and (4) as shown:

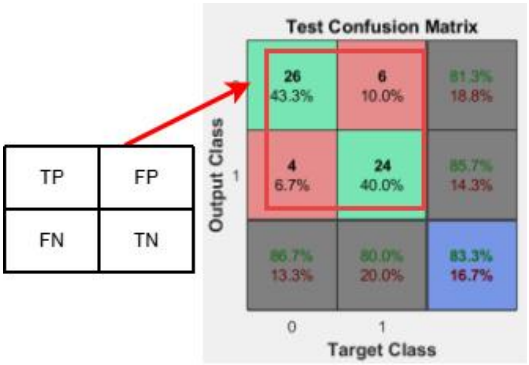


Figure 7: Sample Test Confusion Matrix from traingda Neuron 21

$$\text{Sensitivity (\%)} = \frac{TP}{(TP+FN)} = \frac{26}{(26+4)} = 86.7\% \quad (1)$$

$$\text{Specificity (\%)} = \frac{TN}{(TN+FP)} = \frac{24}{(24+6)} = 80.0\% \quad (2)$$

$$\text{Accuracy (\%)} = \frac{(TN+TP)}{(TN+TP+FN+FP)} = \frac{(24+26)}{(24+26+4+6)} = 83.3\% \quad (3)$$

$$NOC = NOI + (NOI \times HLS) + (HLS \times NOO) \quad (4)$$
$$NOC = 2 + (2 \times 21) + (21 \times 1) = 65$$

Accuracy is the percentage of correct classifications, specificity measures the model's capacity for correctly reject samples unconditionally, and sensitivity evaluates the model's capacity to accurately detect ripe samples. Since the diagnostic model's main objective is correctly identify ripe watermelons, sensitivity and accuracy are given priority in this study. For (4), Number of Connection Counts (NOC) is used to measure the complexity number of connections between the input, hidden, and output layers of a neural network.

III. Results and Discussion

A. Basic Descriptive Statistic

Table V below displays the findings of the basic descriptive statistical analysis. This analysis shows that the VIS and NIR parameters have different ranges of minimum, maximum, mean, and standard deviation values, indicating that there is variation in the optical reflectance values of yellow-flesh watermelon samples. This variation allows the ANN model to learn data patterns more effectively to distinguish maturity grades. Therefore, the descriptive statistical analysis collected spectral data is suitable for use as input in the development of ANN classification models.

TABLE V. Test Of Basic Descriptive Analysis

Input	N Statistic	Min	Max	Mean	Std. Deviation
VIS A	200	47.16	81.53	62.8213	6.44033
VIS B	200	61.60	86.20	71.4688	4.97269
NIR A	200	27.78	57.57	44.2497	6.09312
NIR B	200	26.40	63.35	42.3273	6.70080

B. Neural Network Pattern Recognition & Performance

Three training functions, traingd, traingdm and traingda, were implemented in MATLAB, and each was trained with 14 different numbers of hidden neurons. The sensitivity, specificity and accuracy were determined for each configuration so that the best network for each training function could be identified.

1) Model 1: Gradient descent (traingd)

Figure 8 displays the ANN model's performance based on the zfunction traingd varies with the number of hidden neurons used. The model achieves optimal performance on 7 hidden neurons with accuracy = 80.0%, sensitivity = 80.0% and for specificity = 80.0%, showing a good balance in classifying both grades of watermelon. After neuron 7, several other configurations produce higher sensitivity values, decreasing the specificity value causes the overall performance to be less balanced. This indicates that increasing the number of hidden neurons does not always enhance model performance but could make the network more complex. without providing an increase in classification accuracy. Therefore, the 2–7–1 configuration is chosen as the optimal architecture for the training function traingd.

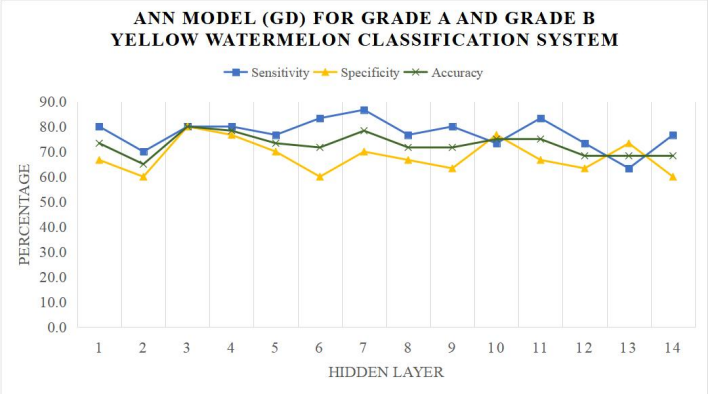


Figure 8: Performance of the traingd model for different numbers of hidden neurons.

2) Model 2: Gradient descent with momentum (traingdm)

Figure 9 shows, the ANN model using the training function traingdm achieved the best performance on 21 hidden neurons with accuracy = 80.0%, sensitivity = 90.0% and specificity = 70.0%. These results indicate that the configuration provides the optimal learning ability for classifying the grade of yellow-fleshed watermelon. After 21 neurons, the model performance did not show a significant improvement but rather tended to decrease. Therefore, the 2–21–1 architecture was selected as the optimal configuration for the training function traingdm.

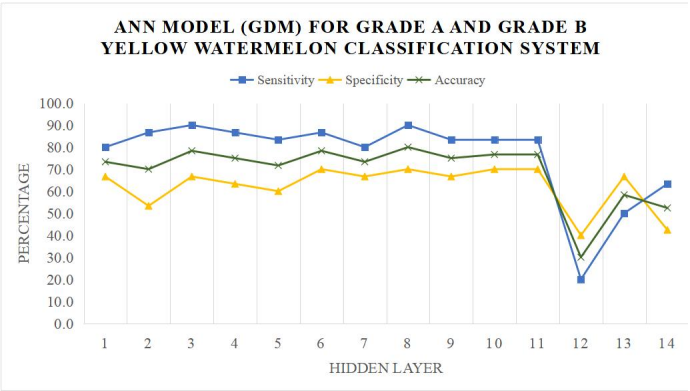


Figure 9: Performance of the traingdm model for different numbers of hidden neurons.

3) Model 3: Gradient descent with adaptive learning rate (traingda)

As seen in Figure 10, the ANN model using the training function traingda achieved the highest performance in the 21 hidden neuron configurations, with accuracy = 83.3%, sensitivity = 86.7% and specificity = 80.0%, indicate that configuration can produce a more balanced classification performance. Although some other neurons showed high sensitivity values, lower accuracy and specificity values caused the overall performance to be less than optimal. Therefore, 2–21–1 is the best configuration for the training function traingda because it provided the best classification performance.

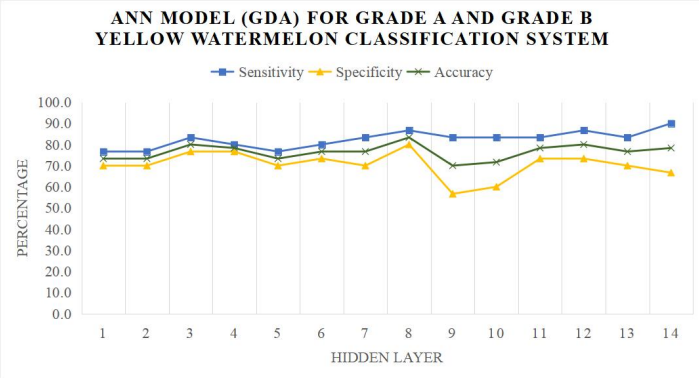


Figure 10: Performance of the traingda model for different numbers of hidden neurons.

C. *Output comparison for all ANN Models*

Based on the number of hidden neurons that produced the best results for each training function, Table VI displays the ideal hidden neuron architecture for each ANN model. The trained model selected 7 hidden neurons by traingd (model 1), creating a 2-7-1 architecture with 23 Number of Connections (NOC). Meanwhile, 21 hidden neurons were selected by the traingdm (model 2) and traingda (model 3), creating a 2-21-1 architecture with 65 NOCs respectively. The difference as shown in Figure 11 indicates that different training algorithms have different learning capacities in determining the ideal number of neurons. The classification performance still depends on the training strategy used throughout the learning process, even though traingdm and traingda use the same number of hidden neurons and NOCs.

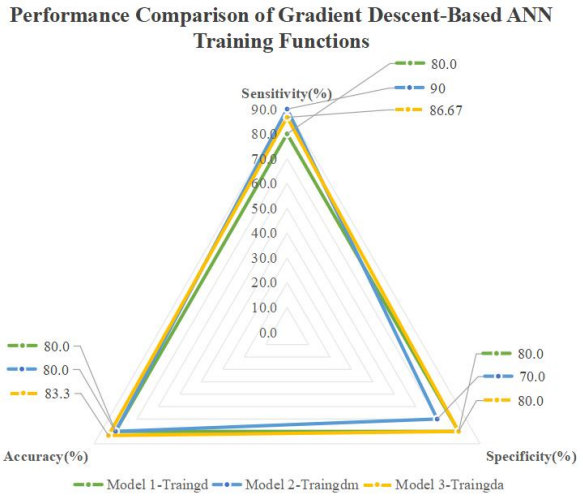


Figure 11: Performance Comparison of the 3 Gradient Descent-Based ANN Training Functions for Yellow-Flesh Watermelon Classification

TABLE VI . COMPARISON BETWEEN 3 MODELS .

No. of Neuron from models	Sensitivity(%)	Specificity(%)	Accuracy(%)	NOC
7 (Model 1)	80.0	80.0	80.0	23
21 (Model 2)	90	70.0	80.0	65
21 (Model 3)	86.67	80.0	83.3	65

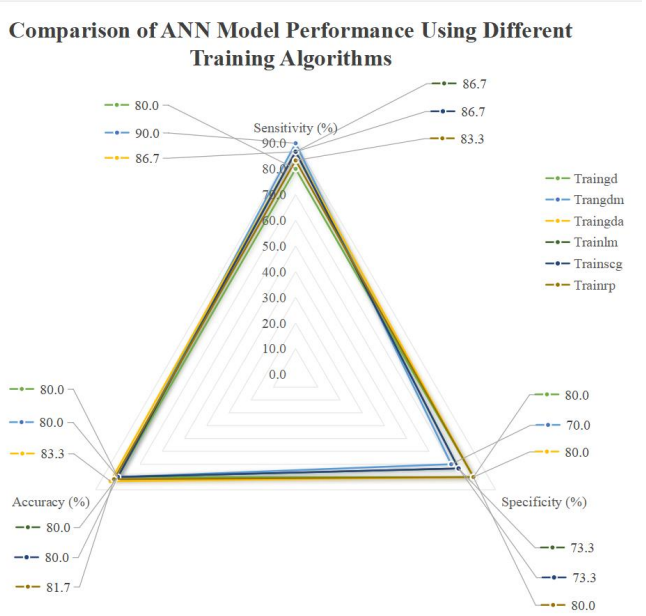


Figure 12: Comparison of the All ANN Training Functions for Yellow-Flesh Watermelon Classification.

TABLE VII . COMPARISON BETWEEN ALL ANN MODELS.

ANN Model Performance	ANIS			AZAMUDDIN		
	Traingd (7)	Traingdm (21)	Traingda (21)	Trainlm (11)	Trainscg (15)	Trainrp (21)
ANN (I/p : Hidden : O/p)	2:7:1	2:21:1	2:21:1	2:11:1	2:15:1	2:21:1
ANN (NOC)	23	65	65	28	36	48
Sensitivity (%)	80.0	90.0	86.7	86.7	86.7	83.3
Specificity (%)	80.0	70.0	80.0	73.3	73.3	80
Accuracy (%)	80.0	80.0	83.3	80	80	81.7

The performance of the ANN models created in this study is compared in Figure 12 and Table VII (traingd, traingdm and traingda) with models from previous studies, namely trainlm, trainscg and trainrp. The results show that each training function produces different performance even though using the same input data. The traingda (model 3) achieved the best overall performance with 83.3% Accuracy, 86.7% Sensitivity and 80.0% Specificity, showing a good balance. The trainrp model recorded the second-best performance with 81.7% Accuracy, but used a 2:21:1 architecture and 48 NOCs, which showed a more complex network than traingda. Meanwhile, trainlm and trainscg each obtained 80.0% Accuracy with 73.3% Specificity, showing satisfactory performance but still lower than traingda.

Although traingdm (model 2) recorded the highest Sensitivity (90.0%), the lower Specificity value (70.0%) caused its overall accuracy not to exceed traingda. In addition,

the traingd (model 1) uses the lowest number of hidden neurons and NOCs, which is 2:7:1 and 23 NOCs but is still able to achieve an Accuracy of 80.0%, indicating that a simpler architecture can also produce good classification performance. Therefore, traingda is selected as the best model because it provides the best balance between Sensitivity, Specificity and Accuracy.

Table VIII provides details for comparing several ANN train function algorithms (GD, GDA, LM, and RP), specifically comparing their advantages and disadvantages to assess their performance and suitability for different applications.

TABLE VIII. ADVANTAGES AND DISADVANTAGES OF MODEL GD, GDA, LM AND RP.

Algorithm	Advantages	Disadvantages
GRADIENT DESCENT (**TRAINGD**)	The algorithm is easy to implement, uses low memory and is suitable for simple ANN networks.	It is easier to become trapped in a local solution and the convergence process is slower..
GRADIENT DESCENT WITH ADAPTIVE LEARNING RATE (**TRAINGDA**)	Automatically modifies the rate of learning, resulting in a more stable training process and improving classification performance.	Requires more hidden neurons and longer training time to achieve optimal performance.
RESILIENT BACKPROPAGATION (**TRAINLM**)	Provides fast convergence and high accuracy for small to medium-sized networks.	Requires high memory usage and risks overfitting if the number of training data is limited.
RESILIENT BACKPROPAGATION (**TRAINRP**)	Using memory efficiently and being less affected by changes in gradient values, increases the stability of the training process.	Less suitable for very complex problems and the choice of training settings determines how well it performs.

IV. CONCLUSION

An ANN-based diagnostic model using VIS and NIR reflectance data successfully demonstrated a comparison between three training functions, showing that each algorithm has different characteristics in terms of network architecture and classification performance. The traingda (model 3) produced the best overall performance with 83.3% Accuracy, 86.7% Sensitivity and 80.0% Specificity, followed by trainrp with 81.7% Accuracy, while traingd (model 1) achieved 80.0% Accuracy using only 7 hidden neurons and 23 Number of Connections (NOC). A lower number of NOCs results in a simpler network architecture, thus reducing the implementation complexity if the model is used on real hardware. On the other hand, models with higher NOCs require more network connections, but potentially provide better classification performance. In terms of training functions, traingd is suitable for applications that require simple models and low resource consumption, traingdm helps speed up the convergence process through the use of momentum, while traingda is more suitable for applications that require high classification accuracy because the level of automatic learning can produce more consistent performance. In addition, comparison with models from previous studies showed that the traingda model provided better classification performance than trainlm and trainscg, and slightly better than trainrp, making it the most suitable model for yellow-fleshed watermelon ripeness classification using VIS/NIR data spectra.

V. FUTURE RECOMMENDATION

In the future, this study can be improved by increasing the number of samples to enhance the model's performance and capacity to adapt for new data. In addition, the application of more spectral features or different VIS/NIR wavelengths has the potential to improve classification accuracy. Further studies can also compare other ANN training algorithms such as trainbr or traincgb, as well as explore CNN - Convolutional Neural Networks and other deep learning techniques to evaluate the potential for improving model performance. In addition, the ANN algorithm also can be combined with other methods such as fuzzy logic, PSO - Particle Swarm Optimization, or GA - genetic algorithms to enhance model performance and optimize network parameters.

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REFERENCES

[1] N. E. Abdullah, N. K. Madzhi, A. M. A. Abg Yahya, A. A. A. Rahim, and A. D. Rosli, "*ANN Diagnostic System for Various Grades of Yellow Flesh Watermelon Based on the Visible Light and NIR Properties*," in Proc. 2018 4th Int. Conf. Electrical, Electronics and System Engineering (ICEESE), Kuala Lumpur, Malaysia, 2018, pp. 70–75, doi: 10.1109/ICEESE.2018.8703498.

[2] E. R. Arboleda, K. M. Parazo, and C. M. Pareja, "*Watermelon ripeness detector using near infrared spectroscopy*," Jurnal Teknologi dan Sistem Komputer, vol. 8, no. 4, pp. 317–322, Oct. 2020, doi: 10.14710/jtsiskom.2020.13744.

[3] R. Balakit, A. J. Castillo, C. J. Garcia, M. C. G. Libang, J. Mallapre, A. Navarro, R. Pangyarihan, K. M. Pariño, E. Wenceslao, and R. A. Reaño, "*Development of Portable Watermelon Ripeness Detector Using Near-*

Infrared Spectroscopy (NIRS) and Acoustics Analysis," PUP Journal of Science and Technology, pp. 36–57, 2023.

[4] D. Jie, W. Zhou, and X. Wei, "*Nondestructive detection of maturity of watermelon by spectral characteristic using NIR diffuse transmittance technique*," Scientia Horticulturae, vol. 257, Art. no. 108718, Nov. 2019, doi: 10.1016/j.scienta.2019.108718.

[5] M. Vega-Castellote, M.-T. Sánchez, I. Torres, M.-J. de la Haba, and D. Pérez-Marín, "*Assessment of watermelon maturity using portable new generation NIR spectrophotometers*," Scientia Horticulturae, vol. 304, Art. no. 111328, Sep. 2022, doi: 10.1016/j.scienta.2022.111328.

[6] A. Ibrahim, H. G. Daood, M. Égei, S. Takács, and L. Helyes, "*A Comparative Study between Vis/NIR Spectroradiometer and NIR Spectroscopy for the Non-Destructive Quality Assay of Different Watermelon Cultivars*," Horticulturae, vol. 8, no. 6, Art. no. 509, Jun. 2022, doi: 10.3390/horticulturae8060509.

[7] D. Jie, L. Xie, X. Rao, and Y. Ying, "*Using visible and near infrared diffuse transmittance technique to predict soluble solids content of watermelon in an on-line detection system,*" Postharvest Biol.Technol, vol. 90, pp. 1–6, Apr. 2014, doi: 10.1016/j.postharvbio.2013.11.009.

[8] X. Sun, S. Xu, and H. Lu, *"Non-Destructive Identification and Estimation of Granulation in Honey Pomelo Using Visible and Near-Infrared Transmittance Spectroscopy Combined with Machine Vision Technology*," Applied Sciences, vol. 10, no. 16, Art. no. 5399, Aug. 2020, doi: 10.3390/app10165399.

[9] H. Tian, Y. Ying, H. Lu, X. Fu, and H. Yu, "*Measurement of soluble solids content in watermelon by Vis/NIR diffuse transmittance technique*," Journal of Zhejiang University Science B, vol. 8, no. 2, pp. 105–110, Feb. 2007, doi: 10.1631/jzus.2007.B0105.

[10] Y. Chen, *"Research on Classification of Watermelon Ripeness Based on Neural Network Pattern Recognition*," in Proc. 2nd International Conference on Computing and Data Science (CONFCDS 2021), Stanford, CA, USA, Jan. 28–30, 2021, pp. 1–5, doi: 10.1145/3448734.3450875.

[11] G. A. de Oliveira, S. Bureau, C. M.-G. C. Renard, A. B. Pereira-Netto, and F. de Castilhos, "*Comparison of NIRS approach for prediction of internal quality traits in three fruit species,*" Food Chemistry, vol. 143, pp. 223–230, Jan. 2014, doi: 10.1016/j.foodchem.2013.07.122.

[12] T. Sun, K. Huang, H. Xu, and Y. Ying, "*Research advances in nondestructive determination of internal quality in watermelon/melon: A review*," J. Food Eng, vol. 100, no. 4, pp. 569–577, Oct. 2010, doi: 10.1016/j.jfoodeng.2010.05.019.

[13] M. F. Mohamedon, F. A. Rahman, S. Y. Mohamad, and O. O. Khalifa, "*Banana Ripeness Classification Using Computer Vision-Based Mobile Application,*" in Proc. 2021 8th International Conference on Computer and Communication Engineering (ICCCE), Kuala Lumpur, Malaysia, Jun. 2021, pp. 335–338, doi: 10.1109/ICCCE50029.2021.9467225.

[14] X. Chen, P. Yuan, and X. Deng, *"Watermelon ripeness detection by wavelet multiresolution decomposition of acoustic impulse response signals*," Postharvest Biol.Technol, vol. 142, pp. 135–141, Aug. 2018, doi: 10.1016/j.postharvbio.2017.08.018.

[15] O. Gupta, A. J. Das, J. Hellerstein, and R. Raskar, "*Machine learning approaches for large scale classification of produce*," Scientific Reports, vol. 8, no. 1, Art. no. 5226, Mar. 2018, doi: 10.1038/s41598-018-23394-3.

[16] M. Rizzo, M. Marcuzzo, A. Zangari, A. Gasparetto, and A. Albarelli, "*Fruit ripeness classification: A survey,*" Artificial Intelligence in Agriculture, vol. 7, pp. 44–57, 2023, doi: 10.1016/j.aiia.2023.02.004.

[17] B. Benmouna, G. García-Mateos, S. Sabzi, R. Fernandez-Beltran, D. Parras-Burgos, and J. M. Molina-Martínez, "*Convolutional Neural Networks for Estimating the Ripening State of Fuji Apples Using Visible and Near-Infrared Spectroscopy*," Food and Bioprocess Technology, vol. 15, no. 10, pp. 2226–2236, Oct. 2022, doi: 10.1007/s11947-022-02880-7.

[18] R. Pourdarbani, S. Sabzi, D. Kalantari, and J. I. Arribas, "*Non-destructive visible and short-wave near-infrared spectroscopic data estimation of various physicochemical properties of Fuji apple (Malus pumila) fruits at different maturation stages*," Chemometrics and Intelligent Laboratory Systems, vol. 206, Art. no. 104147, Dec. 2020, doi: 10.1016/j.chemolab.2020.104147.

[19] O. Elsherbiny, S. El-Hendawy, S. Elsayed, A. E. Elwakeel, A. Alebidi, X. Yue, W. M. Elmessery, and H. Galal, "*Incorporation of visible/near-infrared spectroscopy and machine learning models for indirect assessment of grape ripening indicators,*" Scientific Reports, vol. 15, Art. no. 12345, Apr. 2025, doi: 10.1038/s41598-024-81694-3.

[20] L. A. Varga, J. Makowski, and A. Zell, "*Measuring the ripeness of fruit with hyperspectral imaging and deep learning*," in Proc. 25th International Conference on Pattern Recognition (ICPR), 2021, pp. 2962–2969, doi: 10.1109/ICPR48806.2021.9412504.

[21] B. Deng, X. Wen, and Z. Gao, "*An Improved CNN-Based Neural Network Model for Fruit Sugar Level Detection,*" arXiv:2311.11120, Nov. 2023. doi: 10.48550/arXiv.2311.11120.

[22] M. S. Morshed, S. Ahmed, T. Ahmed, M. U. Islam, and A. B. M. A. Rahman, "*Fruit Quality Assessment with Densely Connected Convolutional Neural Network*," in Proc. 2022 12th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, Dec. 2022, pp. 461–464, doi: 10.1109/ICECE57408.2022.10088873.

[23] Y. Fu, M. Nguyen, and W. Q. Yan, "*Grading Methods for Fruit Freshness Based on Deep Learning,*" SN Computer Science, vol. 3, no. 264, pp. 1–13, Apr. 2022, doi: 10.1007/s42979-022-01152-7.